Relative asymptotic optimality of finite Hermite Gabor windows

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Abstract

The canonical Zak normalization of the positive finite-Hermite ground vector need not minimize the critical orthonormal-Gabor difference energy in a fixed dimension. Nevertheless, it is relatively asymptotically optimal along odd square dimensions. With lattice side length d and the full squared-edge normalization, its rescaled energy is $2\pi \log d + O(1)$, while the minimum is bounded below by $2\pi \log d - 7\pi \log\log d - O(1)$. The quotient of these energies therefore tends to one. The lower bound is proved directly for arbitrary admissible Zak phases, including configurations with multiple defects: small-angle interpolation transfers the discrete energy to growing circular annuli, and the twisted boundary condition forces total integer charge one. A fourth-order oscillator approximation transfers the Gaussian Zak-phase energy to the finite-Hermite candidate. All logarithms are natural.

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1 Main theorem

Let d be a positive integer, $N = d^2$, and equip $\mathbb{C}^N$ with its Euclidean norm. Define

$$(T_N f)(j) = f(j+1), \qquad D_N = I - T_N, \qquad (\mathcal{F}_N f)(k) = N^{-1/2} \sum_{j=0}^{N-1} e^{2\pi i jk/N} f(j),$$

with indices modulo N. Put

$$G_N = D_N^* D_N + \mathcal{F}_N^* D_N^* D_N \mathcal{F}_N, \qquad E_N(f) = \langle f, G_N f \rangle, \qquad A_d(f) = d^2 E_{d^2}(f).$$

The admissible class consists of all vectors f for which

$$\{T_N^{nd} M_N^{md} f : \ 0 \le n, m < d\}, \qquad (M_N f)(j) = e^{2\pi i j/N} f(j),$$

is an orthonormal basis of $\mathbb{C}^N$. There is no additional Fourier-eigenvector condition. Write $a_d = \min_f A_d(f)$ over this class. Multiplying the energy by d^2 does not alter the optimizer at a fixed dimension; it is essential when stating its asymptotic size.

Let $g_{0,d}$ denote the strictly positive Euclidean unit ground vector of G_N, and define

$$Z_N = \mathcal{F}_d \otimes I_d, \qquad \mathcal{J}(v)(j) = \frac{v(j)}{d\,|v(j)|}, \qquad f_d = Z_N^* \mathcal{J}(Z_N g_{0,d}),$$

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when all entries of $Z_N g_{0,d}$ are nonzero. The ground vector is unique up to phase: in standard coordinates,

$$(G_N f)(j) = \big(4 - 2\cos(2\pi j/N)\big) f(j) - f(j-1) - f(j+1),$$

so the Perron–Frobenius theorem applied to a sufficiently large scalar multiple of the identity minus G_N gives a simple positive ground state.

Theorem 1.1. *For every sufficiently large odd d, the vector f_d is well-defined and admissible. As d tends to infinity through odd integers,*

$$A_d(f_d) = 2\pi \log d + O(1),$$

and there is a constant C independent of d such that

$$2\pi \log d - 7\pi \log\log d - C \;\le\; a_d \;\le\; 2\pi \log d + C.$$

Consequently,

$$\frac{E_{d^2}(f_d)}{\min_f E_{d^2}(f)} = \frac{A_d(f_d)}{a_d} \longrightarrow 1.$$

More precisely,

$$0 \le A_d(f_d) - a_d \le 7\pi \log\log d + O(1),$$

and the relative error is $O(\log\log d/\log d)$. The coefficient seven in the lower-order bound is a byproduct of the cutoffs, not a claimed optimal constant. No explicit starting dimension is supplied.

The theorem concerns the natural unitary interpretation of the finite Fourier and Zak transforms. Reversing the Fourier sign consistently only permutes Zak coordinates and does not change the candidate, admissibility, or energy. The original Lammers–Stampe paper [1] has normalization and sign inconsistencies in its displayed formulas; the companion source audit records these issues rather than treating an unnormalized Fourier matrix as unitary.

2 The Zak formulation

In coordinates $j = qd + r$,

$$(Z_N f)(q,r) = d^{-1/2} \sum_{b=0}^{d-1} e^{2\pi i q b/d} f(bd + r).$$

The critical orthonormality criterion is $|(Z_N f)(q,r)| = 1/d$. Thus every admissible vector is uniquely represented by an arbitrary unit-modulus array $\zeta = dZ_N f$ on a $d \times d$ fundamental square. Extend it by

$$\zeta(q+d,r) = \zeta(q,r), \qquad \zeta(q,r+d) = e^{-2\pi i q/d}\zeta(q,r).$$

These are gluing rules for the extension, not extra restrictions on the d^2 free phases. Direct conjugation of T_N and M_N by Z_N gives the exact identity

$$\begin{aligned} A_d(\zeta) &:= d^2 E_{d^2}\big(Z_N^*(\zeta/d)\big) \\ &= \sum_{q,r=0}^{d-1} \Big(|\zeta(q,r+1) - \zeta(q,r)|^2 + |e^{2\pi i r/d^2}\zeta(q+1,r) - \zeta(q,r)|^2 \Big). \end{aligned}$$

Every positive-direction edge occurs once, including both seams; there is no factor $1/2$. The phase torus is compact, so the minimum a_d is attained. The proof first obtains a sharp trial energy and then a matching leading lower bound valid for every array with logarithmic energy.

The Appendix proves the oscillator and Gaussian-Zak estimates used below. In particular, with $h = d^{-1}$, $h_d = \sqrt{d}\, g_{0,d}$ and

$$Z_d b(m,n) = \sum_{r=0}^{d-1} b(m - dr) e^{2\pi i n r/d},$$

one has the uniform approximation (A9), and the Gaussian Zak transform has exactly one simple conformal zero, as in (A10). These facts are proved here, not assumed as additional hypotheses.

3 Candidate energy

3.1 Exact discrete gauge and the Gaussian trial

Write $h = d^{-1}$, $x_m = mh$, $y_n = nh$, and

$$F(x,y) = Z\phi(x,y) = \sum_{r\in\mathbb{Z}} \phi(x-r) e^{2\pi i r y}, \qquad \phi(x) = 2^{1/4} e^{-\pi x^2}.$$

The Schwartz sum is smooth, and its exact gluing laws are

$$F(x+1,y) = e^{2\pi i y} F(x,y), \qquad F(x,y+1) = F(x,y). \tag{U1}$$

The Appendix establishes that the only zero in the unit square is $z_0 = (1/2, 1/2)$, that the zero is simple and conformal in the specified coordinates,

$$F\big(z_0 + (s,t)\big) = a(s+it) + O(s^2+t^2), \qquad a = \partial_x F(z_0) \neq 0, \tag{U2}$$

and that $|F(x,y)| \geq c\,|(x,y) - z_0|$ on the closed fundamental square, after choosing $c > 0$. For odd d, the grid misses the zero. Define the Gaussian phase samples

$$U_{m,n} = \frac{F(x_m, y_n)}{|F(x_m, y_n)|}, \qquad 0 \leq m, n < d,$$

extended off the grid square by $U_{m+d,n} = e^{2\pi i n/d} U_{m,n}$, $U_{m,n+d} = U_{m,n}$. This extension is consistent at corners and follows from (U1).

To record explicitly the relation to the Zak energy above, put

$$W_{m,n} = \frac{Z_d h_d(m,n)}{|Z_d h_d(m,n)|},$$

with the same gluing. The established identity $Z_d h_d(m,n) = d(Z_N g_0)(-n \bmod d,\, m)$ means that the paper-coordinate unit phase $\zeta(q,r) = d\mathcal{J}(Z_N g_0)(q,r)$ satisfies $W_{m,n} = \zeta(-n \bmod d,\, m)$. Reindexing the full Zak energy, including its boundary edges, gives for either $v = U$ or $v = W$ the quadratic expression

$$\mathcal{E}_h(v) = \sum_{m,n=0}^{d-1} \Big(|v_{m+1,n} - v_{m,n}|^2 + |v_{m,n+1} - e^{2\pi i m/d^2} v_{m,n}|^2 \Big), \tag{U3}$$

where values at $m = d$, $n = d$ use the above gluing. Indeed an increasing r-edge is an increasing m-edge, while a q-edge from q to $q+1$, reindexed by $n = -q-1$, becomes the second edge in (U3). Its magnetic multiplier is $e^{2\pi i r/d^2} = e^{2\pi i m/d^2}$. In particular $A_d(f_d) = \mathcal{E}_h(W)$. Formula (U3) is the discrete covariant derivative in the fixed gauge $\mathsf{A} = 2\pi x\, dy$, without a factor $1/2$.

3.2 A lattice vortex estimate with its normalization

Proposition 3.1.

$$\mathcal{E}_h(U) = 2\pi \log(1/h) + O(1). \tag{U4}$$

Proof. All implied constants below are independent of odd d. Choose fixed $0 < R < 1/8$. In the disk $B_{2R}(z_0)$, write $z = s + it$, $r = |z|$, and $V(z) = z/|z|$. Equation (U2), plus the corresponding first-derivative Taylor estimate, implies that

$$H(z) = \frac{F(z_0 + z)}{az} = 1 + O(r), \qquad |\nabla H(z)| \le C \quad (0 < r < 2R). \tag{U5}$$

For example, writing $F = az + G$, one has $G = O(r^2)$, $\nabla G = O(r)$; differentiating $G/(az)$ proves (U5). Shrink R so $|H| \ge 1/2$, and let $B = H/|H|$, $e^{i\alpha} = a/|a|$. Then

$$e^{-i\alpha}\frac{F(z_0 + z)}{|F(z_0 + z)|} = V(z)B(z) = V(z) + R_*(z), \qquad |R_*(z)| \le Cr, \quad |\nabla R_*(z)| \le C. \tag{U6}$$

The last bound uses $|\nabla V| \le 1/r$. No smooth extension of R_* through the origin is asserted or needed.

For a grid basepoint with $4h \le r_{m,n} := |(x_m, y_n) - z_0| < R$, each forward edge segment remains at distance at least $3r_{m,n}/4$ from z_0 and lies in B_{2R} for small h. Hence (U6) gives an $O(h)$ difference between the ordinary U-edge increment (after removing the harmless constant $e^{i\alpha}$) and the corresponding V-edge increment. The magnetic multiplier in the y-edge alters the former by at most $|e^{2\pi i m/d^2} - 1| \le 2\pi h$. Since $|V(z + he_j) - V(z)| \le Ch/r_{m,n}$, the difference of *squared* edge lengths is bounded by

$$C\Big(\frac{h^2}{r_{m,n}} + h^2\Big). \tag{U7}$$

Summed over these basepoints and both directions, (U7) is $O(1)$. One direct count partitions the disk into dyadic shells $r \asymp \rho_k = 2^k h$: a shell has at most $C(\rho_k/h)^2$ grid sites, so its contribution from h^2/r is at most $C\rho_k$, and $\sum_{\rho_k \le R} \rho_k = O(R)$. The h^2 terms sum to $O(1)$.

There are only $O(1)$ sites with $r_{m,n} < 4h$, and each squared edge of unit phases is at most 4; these core edges therefore cost $O(1)$, whether or not an edge approaches the zero closely. On basepoints outside $B_R(z_0)$, the phase $F/|F|$ has a uniformly bounded derivative along all forward edge segments for small h; the only seams are identified by the smooth, exact gluing (U1). Each ordinary increment is $O(h)$, and the y-magnetic correction is also $O(h)$. There are $O(h^{-2})$ such sites, so all their squared edges total $O(1)$. We have reduced (U4), up to $O(1)$, to the *unmagnetized* V-edge energy with $4h \le r_{m,n} < R$.

For those sites Taylor's theorem along an edge, using $|\nabla V| \le C/r$, $|\nabla^2 V| \le C/r^2$, gives

$$\Big| |V(z + he_j) - V(z)|^2 - h^2 |\partial_j V(z)|^2 \Big| \le C\frac{h^3}{r^3}. \tag{U8}$$

The same shell count makes the sum of (U8) $O(1)$: on a shell of radius ρ_k, its bound is $Ch/\rho_k = C2^{-k}$. Also

$$|\partial_s V|^2 + |\partial_t V|^2 = \frac{1}{r^2}. \tag{U9}$$

Consequently the vortex edge energy is

$$\sum_{4h \le r_{m,n} < R} \frac{h^2}{r_{m,n}^2} + O(1).$$

Here is a direct singular Riemann-sum justification rather than an appeal to an ordinary smooth-function theorem. Associate to each included grid point its square $Q_{m,n}$ of side h centered there. As $r_{m,n} \geq 4h$, the mean-value bound $|\nabla(r^{-2})| \leq Cr^{-3}$ on its square shows

$$\Big| \int_{Q_{m,n}} \frac{dw}{|w - z_0|^2} - \frac{h^2}{r_{m,n}^2} \Big| \leq C \frac{h^3}{r_{m,n}^3}.$$

The sum of these errors is $O(1)$ by the same shells. The union of the squares differs from the annulus $4h \leq r < R$ only in strips of width $O(h)$ around its two circles. The integral of r^{-2} over the inner strip is $O(1)$, and over the outer strip is $O(h/R)$. Thus

$$\sum_{4h \leq r_{m,n} < R} \frac{h^2}{r_{m,n}^2} = \int_{4h \leq r < R} \frac{ds\, dt}{r^2} + O(1) = 2\pi \log \frac{R}{4h} + O(1) = 2\pi \log(1/h) + O(1). \tag{U10}$$

Equations (U7)–(U10) prove (U4). The angular integral is $\int_0^{2\pi} d\theta = 2\pi$. The coefficient would be π only for an energy with an extra $1/2$, which the present energy does not contain. □

3.3 Quantitative transfer to the finite-Hermite phase

Use the uniform estimate (A9) from the Appendix:

$$Z_d h_d(m,n) = c_d \big\{ F(x_m, y_n) + h^2 Z\psi(x_m, y_n) \big\} + O(h^3), \qquad c_d > 0,\ c_d \to 1, \tag{U11}$$

with a grid-uniform remainder. The fixed Schwartz function ψ has bounded Zak transform. Dividing (U11) by the positive c_d leaves its phase unchanged and gives $\sup_{m,n} |c_d^{-1} Z_d h_d - F(x_m, y_n)| \leq Ch^2$ for sufficiently small h. The Appendix's nonvanishing estimate ensures W exists for all sufficiently large odd d. The elementary normalization inequality $|p/|p| - q/|q|| \leq 2|p-q|/|q|$, and $|F| \geq cr$, now give

$$|W_{m,n} - U_{m,n}| \leq \frac{Ch^2}{r_{m,n}}. \tag{U12}$$

The half-offset odd grid has $r_{m,n} \geq h/\sqrt{2}$. Dyadic counting as above, up to a disk containing the unit square, yields

$$\sum_{m,n} \frac{h^2}{r_{m,n}^2} = O(\log(1/h)), \qquad \|W - U\|_{\ell^2(\{0,\dots,d-1\}^2)} = O\big(h\sqrt{\log(1/h)}\big). \tag{U13}$$

The gluing of W is exactly that of U, because the discrete Zak transform has the same identities (U1) on lattice translates. Thus $\delta = W - U$ is a valid twisted array in the same Hilbert space. Let D_h be the pair of covariant forward differences in (U3). Its two weighted shifts are unitary (including the twisted boundary shift), so $\|D_h \delta\|_2 \leq 2\sqrt{2} \|\delta\|_2$. By (U4), $\|D_h U\|_2 = O(\sqrt{\log(1/h)})$. The polarization/Cauchy–Schwarz estimate therefore gives

$$|\mathcal{E}_h(W) - \mathcal{E}_h(U)| \leq 2\|D_h U\|_2 \|D_h \delta\|_2 + \|D_h \delta\|_2^2 = O(h \log(1/h)) = o(1). \tag{U14}$$

Combining (U4) and (U14) proves the claimed asymptotic for $A_d(f_d)$. In particular the uniform $O(h^2)$ *unnormalized* Zak approximation is more than sufficient to transfer the total energy by $O(1)$, even by $o(1)$. The phase error itself is not uniformly $O(h^2)$ at the zero: (U12) correctly allows $O(h)$ on nearest sites; its summed norm is what makes (U14) work. This proof uses the exact twisted seam in both arrays, rather than silently treating a nonperiodic phase as periodic.

4 Universal lower bound

Let $d \geq 3$, $|\zeta(q,r)| = 1$, and impose

$$\zeta(q+d,r) = \zeta(q,r), \qquad \zeta(q,r+d) = e^{-2\pi i q/d}\zeta(q,r).$$

These identities specify the lift of the array to $\mathbb{Z}^2$, including the edges crossing the seams. Put

$$a^q_{q,r} = \operatorname{Arg}\Big(e^{2\pi i r/d^2}\zeta(q+1,r)\overline{\zeta(q,r)}\Big), \qquad a^r_{q,r} = \operatorname{Arg}\Big(\zeta(q,r+1)\overline{\zeta(q,r)}\Big)$$

with principal arguments in $(-\pi,\pi]$. These are periodic modulo d, despite the twist in ζ. The Zak energy above, counting every positive-oriented edge *once* and *without a factor* $1/2$, is

$$A_d(\zeta) = \sum_{q,r \pmod d} \big(4\sin^2(a^q_{q,r}/2) + 4\sin^2(a^r_{q,r}/2)\big). \tag{L1}$$

Theorem 4.1. *Assume merely that a trial array gives* $\min A_d = O(\log d)$. *Then*

$$\liminf_{d\to\infty} \frac{\min_{|\zeta|=1} A_d(\zeta)}{\log d} \;\geq\; 2\pi. \tag{L2}$$

The proof applies to odd or even d. More precisely, every family with $A_d(\zeta_d) \leq C\log d$, with fixed C, satisfies $A_d(\zeta_d) \geq 2\pi\log d - 7\pi\log\log d - O_C(1)$. No assumption is made on the number, locations, or signs of its defects.

The proof occupies the rest of this section.

4.1 Exact discrete topological identity

Orient a plaquette in the order $+q, +r, -q, -r$. Its oriented sum of principal covariant differences has the form

$$a^q_{q,r} + a^r_{q+1,r} - a^q_{q,r+1} - a^r_{q,r} = -\frac{2\pi}{d^2} + 2\pi n_{q,r}, \qquad n_{q,r} \in \mathbb{Z}. \tag{L3}$$

Indeed, the product of the four underlying unit complex numbers is $e^{-2\pi i/d^2}$. The seam edges obey the same identity when the indicated lifts of ζ are used. Summing (L3) over all d^2 plaquettes cancels *every* oriented edge, since the covariant edge numbers are periodic, and gives

$$\sum_{q,r} n_{q,r} = 1. \tag{L4}$$

In particular, there is always a nonzero integer defect somewhere. This formulation already handles the nontrivial torus bundle: no globally periodic scalar phase has been assumed.

Choose $\eta_d = (\log d)^{-1}$ (for sufficiently large d so $\eta_d < \pi/4$). Call an edge *large* if $|a_e| \geq \eta_d$, and a plaquette *bad* if it is incident to a large edge. From (L1),

$$\#\{\text{large edges}\} \leq \frac{A_d}{4\sin^2(\eta_d/2)} \leq C_1\frac{\log d}{\eta_d^2}, \qquad \#\{\text{bad plaquettes}\} \leq 2\#\{\text{large edges}\}. \tag{L5}$$

For every plaquette that is not bad, all four differences have magnitude less than η_d. Equation (L3) then forces $n_{q,r} = 0$, since $4\eta_d + 2\pi/d^2 < 2\pi$. Thus *all* nonzero defects are among the $O((\log d)^3)$ bad plaquettes. We cover *all* bad plaquettes, including those with $n = 0$, rather than assume only one defect or attempt to discard expensive neutral cores.

4.2 Local interpolation and the degree lemma

Identify the discrete torus with unit squares on the flat torus $\mathbb{R}^2/d\mathbb{Z}^2$, plaquette (q, r) being $[q, q+1] \times [r, r+1]$. Let B be any Euclidean disk of radius at most $R_d + O(1)$, where

$$R_d = \frac{d}{\sqrt{\log d}} = o(d). \tag{L6}$$

For large d, this disk, with a two-unit collar, has a unique embedded lift to the plane after choosing a lift of its center (q_0, r_0). On it use the gauge

$$\zeta'(q, r) = e^{2\pi i q r_0/d^2} \zeta(q, r).$$

In this gauge the covariant link angles are represented by $A'_q(q, r) = 2\pi(r - r_0)/d^2$, $A'_r(q, r) = 0$. For every edge touching this lifted disk,

$$|A'_e| \le C \frac{R_d + O(1)}{d^2} =: \delta_d. \tag{L7}$$

The gauge is only asserted on this *local lift*, not globally on the torus. If an edge belongs to a good square, let $b_e = \mathrm{Arg}(\zeta'_{\mathrm{head}} \overline{\zeta'_{\mathrm{tail}}})$. Since $|a_e| < \eta_d$ and $\delta_d \to 0$, the principal-branch identity is exactly $b_e = a_e - A'_e$. Summing around a good square and using (L3) and the circulation $-2\pi/d^2$ of A' gives $\sum_{\partial p} b_e = 2\pi n_p = 0$. Consequently its four vertex phases have a consistent real lift. Bilinearly interpolate that real phase across the square and exponentiate. Interpolants on adjacent good squares agree on their common edge, since there they interpolate the same principal phase difference b_e; they thus give an H^1, S^1-valued map on any union of good squares in the disk. No lift across a bad square is needed.

The following *pointwise* estimate avoids paying for a square more than once when disks or gauges change during ball growth. For a good unit square, write $a_{\mathrm{bottom}}, a_{\mathrm{top}}, a_{\mathrm{left}}, a_{\mathrm{right}}$ for its four covariant edge numbers, and at local coordinates $(x, y) \in [0, 1]^2$ define

$$W(x, y) = (1 - y) a^2_{\mathrm{bottom}} + y a^2_{\mathrm{top}} + (1 - x) a^2_{\mathrm{left}} + x a^2_{\mathrm{right}}. \tag{L8}$$

For the bilinear phase θ, its derivatives are convex combinations of the corresponding b's. Jensen's inequality and $(s - t)^2 \le (1 + \eta_d) s^2 + (1 + \eta_d^{-1}) t^2$ therefore give, in *any* of the above local gauges,

$$|\nabla\theta(x, y)|^2 \le (1 + \eta_d) W(x, y) + C \eta_d^{-1} \delta_d^2. \tag{L9}$$

The function W uses the periodic, gauge-invariantly defined covariant numbers a_e, so it is a **single common weight** independent of the ball or gauge. Integrating (L8) over all good squares counts any edge at most twice with coefficient $1/2$. Also, for a good edge, $a_e^2 \le (1 + C\eta_d^2) 4 \sin^2(a_e/2)$. Hence

$$\sum_{p \text{ good}} \int_p W \le (1 + C\eta_d^2) A_d. \tag{L10}$$

Even if the union of swept annuli touches all d^2 squares, the integrated last term of (L9) is at most

$$C \eta_d^{-1} d^2 \delta_d^2 \le C' \eta_d^{-1} \Big(\frac{R_d + O(1)}{d} \Big)^2 = O(1). \tag{L11}$$

Lemma 4.2 (Cellular degree lemma)**.** *Let $\mathcal{C} = \partial B(c, \rho)$ be the boundary of one ball during a collision-free growth interval. Suppose a collar of width greater than two around $\mathcal{C}$ contains no bad plaquette. Let $Q(\mathcal{C})$ be the sum of n_p over bad plaquettes inside $\mathcal{C}$. Then the interpolated map in the local gauge is defined on a neighborhood of $\mathcal{C}$, and*

$$\deg(u|_{\mathcal{C}}) = Q(\mathcal{C}). \tag{L12}$$

Proof. Form a cellular two-chain $D_{\mathcal{C}}$ from all unit plaquettes whose centers lie in $B(c, \rho)$, with a harmless consistent tie convention if a center lies on the circle. If an edge belongs to the cellular boundary $\partial D_{\mathcal{C}}$, the centers of its two incident plaquettes lie on opposite sides of the circle. Hence that edge, and the two incident plaquettes, lie within a fixed distance less than two of $\mathcal{C}$. They are good by the collar hypothesis. The symmetric difference between the geometric disk and $D_{\mathcal{C}}$ is contained in the same good collar. Therefore the oriented one-cycle $\partial D_{\mathcal{C}}$, possibly with several components, is homologous to $\mathcal{C}$ within a region on which the interpolated S^1 map is defined. Degree is additive over the components of an oriented cycle, so

$$2\pi \deg(u|_{\mathcal{C}}) = \sum_{e \subset \partial D_{\mathcal{C}}} b_e.$$

Discrete Stokes, including all interior bad edges, gives

$$\begin{aligned}
\sum_{e \subset \partial D_{\mathcal{C}}} b_e &= \sum_{e \subset \partial D_{\mathcal{C}}} (a_e - A'_e) \\
&= \Big(-\frac{2\pi \# D_{\mathcal{C}}}{d^2} + 2\pi \sum_{p \in D_{\mathcal{C}}} n_p\Big) - \Big(-\frac{2\pi \# D_{\mathcal{C}}}{d^2}\Big) \\
&= 2\pi \sum_{p \in D_{\mathcal{C}}} n_p.
\end{aligned}$$

All nonzero n_p occur on bad plaquettes, and the collar ensures that $D_{\mathcal{C}}$ and the geometric disk contain the same bad plaquettes. Thus the last sum is $2\pi Q(\mathcal{C})$. If a circle meets a grid vertex or a tie occurs, perturb ρ slightly inside the defect-free collar and pass back by homotopy; alternatively the argument is needed only for almost every ρ, which already suffices for the polar integral. □

For a complete Euclidean annulus $r < |x - c| < s$ swept by a ball of integer charge Q, (L12) and Cauchy–Schwarz on every circle give the **exact continuum coefficient**

$$\int_{r<|x-c|<s} |\nabla\theta|^2 \, dx \;\geq\; \int_r^s \frac{1}{\rho} \frac{(2\pi Q)^2}{2\pi} \, d\rho = 2\pi Q^2 \log \frac{s}{r}. \tag{L13}$$

The polar derivative here is the derivative of a local real lift of the S^1 map along the circle. Its total increment is $2\pi Q$. In particular, (L13) has coefficient 2π, not π: equation (L1) contains the full squared-edge energy, not half of it. Using square lattice contours directly would lose sharpness through their anisotropic perimeter; bilinear interpolation is what recovers circular contours without an energy penalty.

4.3 Ball growth with arbitrary defects and merges

Here are the elementary geometric details used in the proof. Place an initial Euclidean ball of radius 10 around the center of **each** bad unit square, on the flat torus. Each bad square lies strictly inside its own ball. Merge intersecting balls until the family is disjoint, replacing two balls of radii r_1, r_2 by a containing ball of radius **exactly** $r_1 + r_2$. Such a ball exists: when the center distance is at most $r_1 + r_2$, choose its center along a shortest segment so its distances from the two old centers are at most r_2, r_1, respectively. At all stages here the radii and their sum are $o(d)$, so the lifted Euclidean construction gives embedded torus disks. A replacement can intersect a third ball; merge again. There are finitely many replacements, and the sum of radii remains

$$S_0 = 10\,\#\{\text{bad plaquettes}\} \le C_2(\log d)^3. \tag{L14}$$

Now grow every ball homothetically at the same rate: in a time interval with no collision, its radius is $r_i(t) = t\, r_i(t_0)/t_0$. At a contact, merge the touching balls in the same way, iterating any resulting overlaps; label the new ball by the sum of the old integer charges. A ball's charge is the sum of n_p over the bad plaquettes assigned to it. All initial bad plaquettes are assigned to their own radius-ten balls.

For every bad-square center x assigned to a current ball $B(c, r)$, maintain

$$r - |x - c| \ge 10.$$

It holds initially. During homothetic growth the left side can only increase. If balls $B(c_1, r_1)$ and $B(c_2, r_2)$ meet, choose a center c on a shortest lifted segment so that

$$|c - c_1| \le r_2, \qquad |c - c_2| \le r_1,$$

and give the replacement radius $r_1 + r_2$. Then, for a center x assigned to the first ball,

$$(r_1 + r_2) - |x - c| \ge r_1 - |x - c_1| \ge 10,$$

and similarly for the second. Recursive merging preserves the invariant. Since a point of a unit square is at distance at most $\sqrt{2}/2$ from its center, every bad square lies more than nine units inside its assigned ball. While the balls are disjoint, a boundary circle of any one ball is also more than nine units from bad squares assigned to every other ball. Thus the claimed two-unit good collar is available uniformly, including just before a collision.

If a newly created ball contains or comes within two units of a bad square assigned to a third ball, it intersects that third ball because the third bad-square center has ten units of inward margin. The prescribed recursive merger therefore absorbs that ball too. After all overlaps are resolved, the balls genuinely partition all bad plaquettes, so their integer labels satisfy

$$\sum_i Q_i = \sum_p n_p = 1.$$

Because the Q_i are integers,

$$\sum_i Q_i^2 \ge 1. \tag{L15}$$

This remains true despite cancellation of positive and negative defects at mergers. A neutral merged ball may contribute zero, but the total charge one cannot disappear. No stronger assertion such as $\sum|Q_i|$ conservation is needed.

The radius sum stays exactly tS_0, including through mergers (using radius equal to the sum, not merely bounded by it). Stop at $T_d = R_d/S_0$; for large d, $T_d > 1$, and every radius at every time is at most $R_d = o(d)$.

Within any collision-free time interval (t_0, t_1), each ball sweeps its full concentric annulus from radius $r_i(t_0)$ to $r_i(t_1)$. For disjointness of swept annuli, let $U(t)$ be the union of current balls. Between collisions the swept annuli are disjoint because the balls are disjoint. At a collision the replacement ball contains the old balls, so $U(t)$ only jumps outward. The jump region is deliberately not counted. Every later swept annulus lies outside the post-collision union, while every earlier swept annulus lies inside it. Hence annuli from different collision-free intervals are disjoint up to boundaries of measure zero. This monotonicity permits use of the common-weight estimate.

No swept annulus meets a bad square, which was inside the initial union. Each annulus therefore lies in good squares and admits the local interpolation and cellular degree lemma above. Use all good squares meeting the annulus, with the gauge defined on its ball enlarged by two units. Squares can meet several annuli, but the annular subsets themselves are disjoint: integrate the common pointwise weight W over those subsets first, and only then enlarge to the union of all good squares. At collision instants the replacement ball may engulf extra territory; this territory is *not* charged to any annulus, which only weakens the estimate. The finite set of collision times has no effect on the integrals. Combining (L13) over all disjoint swept annuli yields

$$\sum_{\text{annuli}} \int |\nabla\theta|^2 \geq 2\pi \int_1^{T_d} \sum_i Q_i^2 \, \frac{dt}{t} \geq 2\pi \log T_d. \tag{L16}$$

Although separate annuli may use different gauges, (L9) bounds each of their densities by the *same* lattice weight W; disjointness, (L10), and (L11) show

$$2\pi \log T_d \leq (1 + C\eta_d) A_d + O(1). \tag{L17}$$

Finally, (L6) and (L14) imply

$$\log T_d = \log d - \tfrac{1}{2} \log\log d - \log S_0 \geq \log d - \tfrac{7}{2} \log\log d - O(1). \tag{L18}$$

Since $A_d \leq C_0 \log d$ and $\eta_d = (\log d)^{-1}$, the multiplicative loss $C\eta_d A_d$ is bounded. Consequently (L17)–(L18) give the quantitative form

$$A_d \geq 2\pi \log d - 7\pi \log\log d - C_{C_0}. \tag{L19}$$

In particular they establish (L2), first for every family with a fixed logarithmic energy bound and then for minimizers under the trial upper bound. This uses only the self-contained interpolation and ball-growth argument, not an external vortex theorem. The cutoff choice also makes the total gauge error in (L11) bounded; it need not tend to zero. This completes the proof of Theorem 4.1.

5 Completion of the proof and interpretation

The candidate estimate gives $a_d \leq A_d(f_d) \leq 2\pi \log d + C_1$ for all sufficiently large odd d. Choose a minimizing array for each such d. These arrays have a common logarithmic energy bound, so (L19) applies and yields

$$a_d \geq 2\pi \log d - 7\pi \log\log d - C_2.$$

Together with the candidate upper estimate, this proves all stated absolute bounds. In particular $a_d \sim 2\pi \log d > 0$, and

$$0 \le \frac{A_d(f_d)}{a_d} - 1 = \frac{A_d(f_d) - a_d}{a_d} \le \frac{7\pi \log\log d + C_3}{2\pi \log d - 7\pi \log\log d - C_2} \longrightarrow 0.$$

This is a statement about the full orthonormal-generator class, not a symmetry-restricted problem. In the original unscaled energy, both the candidate and the minimum are asymptotic to $2\pi \log d/d^2$. This proves Theorem 1.1.

Relative optimality does not restore exact or local optimality. A fixed positive improvement in A_d is compatible with a quotient tending to one, because the optimal energy itself diverges logarithmically. The companion eventual-nonoptimality note supplies such a one-coordinate descent for all sufficiently large odd d. The present argument does not show that the optimality gap converges, remains bounded, or equals that particular one-site improvement; it proves only the displayed $O(\log\log d)$ upper bound on the gap. Nor does it establish uniqueness of asymptotically optimal windows.

The coefficient 2π is the energy of one unit phase winding in the full Dirichlet normalization. The role of the Hermite construction is to realize that coefficient with a controlled core and a smooth exterior phase. The universal obstruction comes from the twisted Zak boundary condition, which forces total integer defect one regardless of how defects of opposite signs are distributed or merged.

A Oscillator approximation and the Gaussian Zak zero

For completeness, the analytic inputs used in the candidate estimate are proved here. In this Appendix, write $g_0 = g_{0,d}$.

A.1 Spectral convergence of the finite oscillator

Put $h = d^{-1}$, and, for odd d, choose the d^2 nodes $x_j = jh$, $-(d^2-1)/2 \le j \le (d^2-1)/2$, on the circle $[-d/2, d/2]$ with identified endpoints. Let $\|b\|_d^2 = h\sum_j |b_j|^2$. The operator $Q_d = d^2 G_{d^2}$ has form

$$\mathfrak{q}_d(b) = \langle b, Q_d b\rangle_d = h\sum_j \Big[h^{-2}|b_{j+1} - b_j|^2 + V_d(x_j)|b_j|^2\Big], \qquad V_d(x) = 4h^{-2}\sin^2(\pi h x),$$

with cyclic differences. In the centered interval, $\sin(\pi|x|/d) \ge 2|x|/d$, hence $V_d(x_j) \ge 16x_j^2$. On bounded sets, $V_d(x_j) \to 4\pi^2 x_j^2$ uniformly. Denote the two lowest eigenvalues of Q_d by $\lambda_{0,d} < \lambda_{1,d}$. Its positive, $\|\cdot\|_d$-unit ground vector is $h_d = \sqrt{d}\, g_0$.

Here is a precise compactness argument, including the growing-circle seam. Let $I_d b$ be periodic linear interpolation between nodes, and $J_d b$ the step function equal to b_j on the centered Voronoi cell $[x_j - h/2, x_j + h/2)$. The last and first nodes are joined across the identified endpoints; both are at distance $h/2$ from an endpoint. If $K_d = h\sum_j h^{-2}|b_{j+1} - b_j|^2$, direct integration on

each edge gives

$$\begin{aligned} \|J_d b\|^2_{L^2(-d/2,d/2)} &= \|b\|_d^2, \\ \int_{-d/2}^{d/2} |(I_d b)'|^2 &= K_d, \\ \|I_d b\|_2^2 &= \|b\|_d^2 - \frac{h^2}{6} K_d, \\ \|I_d b - J_d b\|_2^2 &= \frac{h^2}{12} K_d. \end{aligned} \tag{A1}$$

These are also valid for complex samples. Since $V_d(x_j) \geq 16x_j^2$, a unit vector with $\mathfrak{q}_d(b) \leq C$ has $K_d \leq C$ and $h\sum_j x_j^2 |b_j|^2 \leq C/16$. On each Voronoi cell $|x - x_j| \leq h/2$, so

$$\int x^2 |J_d b(x)|^2 \, dx \leq 2h \sum_j x_j^2 |b_j|^2 + \frac{h^2}{2} \|b\|_d^2 \leq C/8 + 1/2. \tag{A2}$$

Extend $J_d b$ by zero outside the circle. Equation (A2) bounds its mass outside $[-R, R]$ by $O_C(R^{-2})$, uniformly in d. Locally, $I_d b$ is uniformly bounded in H^1, hence precompact in L^2 on every fixed interval; (A1) transfers this compactness to $J_d b$. Diagonal extraction plus (A2) therefore gives strong *global* $L^2(\mathbb{R})$ convergence of a subsequence of the step functions. In particular, norms and mutual inner products of any fixed finite family of bounded-energy vectors survive in the limit. On each bounded interval, the interpolants converge weakly in H^1; the step potentials $W_d(x) = V_d(x_j)$ on the cells converge locally uniformly to $4\pi^2 x^2$. Lower semicontinuity and local strong L^2 convergence imply, for a limit u,

$$\mathfrak{q}_0(u) := \int_{\mathbb{R}} \left(|u'|^2 + 4\pi^2 x^2 |u|^2 \right) dx \leq \liminf_{d \to \infty} \mathfrak{q}_d(b_d). \tag{A3}$$

To obtain (A3), first use bounded intervals and then let their radius grow. Thus no mass can escape through the seam or to infinity.

For the matching upper bounds, take Schwartz functions f and their periodized samples

$$S_d f(j) = \sum_{k \in \mathbb{Z}} f(x_j + kd).$$

Schwartz decay and Riemann sums give $\langle S_d f, S_d g \rangle_d \to \langle f, g \rangle_{L^2}$; Taylor expansion, or the operator estimate in the next subsection, gives $\langle S_d f, Q_d S_d g \rangle_d \to \langle f, H_0 g \rangle$, where $H_0 = -\partial_x^2 + 4\pi^2 x^2$. This holds for any fixed pair of Schwartz functions. The Gaussian $\phi = 2^{1/4} e^{-\pi x^2}$ has norm one and eigenvalue 2π, and $x\phi$ is an orthogonal eigenfunction with eigenvalue 6π. Their sampled two-dimensional span gives $\limsup \lambda_{0,d} \leq 2\pi$ and $\limsup \lambda_{1,d} \leq 6\pi$ by min–max.

For completeness, H_0 has exactly these first two eigenvalues. Its form domain embeds compactly in L^2, by local H^1 compactness and the x^2 tail bound, so min–max applies. With $a = \partial_x + 2\pi x$, $H_0 = a^* a + 2\pi$; the kernel of a is precisely $\mathbb{C}\phi$. The identity $H_0 a = a(H_0 - 4\pi)$ shows an eigenfunction with eigenvalue strictly between 2π and 6π would give a nonzero L^2 distributional eigenfunction of eigenvalue less than 2π, which is impossible. To justify that last assertion without an unmentioned decay hypothesis, if $w \in L^2$ solves $(H_0 - \nu)w = 0$ distributionally, local ODE regularity and a compactly supported cutoff χ_R yield

$$\int \left(|(\chi_R w)'|^2 + 4\pi^2 x^2 |\chi_R w|^2 \right) = \nu \|\chi_R w\|_2^2 + \int |\chi_R'|^2 |w|^2.$$

The factorization lower bound $2\pi\|\chi_R w\|_2^2$ on the left and $\|\chi_R'\|_\infty = O(R^{-1})$ force $w = 0$ if $\nu < 2\pi$. Finally $a^*\phi = 4\pi x\phi$ has eigenvalue 6π. Applying (A3) to normalized first and second discrete eigenvectors, simultaneously extracting their two orthogonal limits, gives $\liminf \lambda_{0,d} \geq 2\pi$ and $\liminf \lambda_{1,d} \geq 6\pi$ by continuum min–max (for the second, every unit combination of the two limits has form at most the limit of $\lambda_{1,d}$). Consequently

$$\lambda_{0,d} \to 2\pi, \qquad \lambda_{1,d} \to 6\pi. \tag{A4}$$

In particular, the gap above the ground state is uniformly positive for sufficiently large odd d.

A.2 A fourth-order approximate ground vector

Write $\varepsilon = h$. For every fixed Schwartz f, one has in sampled norm

$$Q_d S_d f = S_d(H_0 f + h^2 A f) + O_{\ell^2_d}(h^4), \qquad Af = -\frac{1}{12} f^{(4)} - \frac{4}{3}\pi^4 x^4 f. \tag{A5}$$

Here the error constant depends on f, not on d. Indeed, periodization commutes exactly with the centered difference and with $V_d(x)$, whose period is d. Taylor's integral remainder for the centered second difference is bounded pointwise by $Ch^4 \sup_{|z-x|\leq h}|f^{(6)}(z)|$; and the global bound

$$\left|4h^{-2}\sin^2(\pi h x) - 4\pi^2 x^2 + \frac{4}{3}\pi^4 h^2 x^4\right| \leq Ch^4|x|^6$$

follows from the order-six remainder for $\sin^2 z = z^2 - z^4/3 + O(z^6)$, valid globally after enlarging C for $|z| \geq 1$. Both resulting remainders are h^4 times functions with a uniform, integrable Schwartz envelope. If $|F_h(x)| \leq C(1+|x|)^{-p}$, $p > 2$, then $\|S_d F_h\|_d \leq C_p$ uniformly: on $|x_j| \leq d/2$, separate the central translate from the sum of other translates, which is $O(d^{-p})$, and sum the squares with weight h. This establishes (A5), including the seam and all-periodized-tail errors.

Put

$$q(x) = \frac{\pi^3 x^4}{6} - \frac{\pi^2 x^2}{4} + \frac{\pi}{32}, \qquad \psi = q\phi, \qquad v_d = S_d(\phi + h^2\psi), \qquad \Lambda_d = 2\pi - \frac{\pi^2 h^2}{2}.$$

Direct differentiation gives

$$A\phi = \left(-\tfrac{8}{3}\pi^4 x^4 + 4\pi^3 x^2 - \pi^2\right)\phi,$$
$$(H_0 - 2\pi)\psi = \left(\tfrac{8}{3}\pi^4 x^4 - 4\pi^3 x^2 + \tfrac{\pi^2}{2}\right)\phi,$$

thus $(H_0 - 2\pi)\psi = -(A + \pi^2/2)\phi$. Applying (A5) separately to ϕ and ψ, and noting that $S_d(A\psi)$ is uniformly bounded, proves

$$\|(Q_d - \Lambda_d)v_d\|_d \leq Ch^4. \tag{A6}$$

The Gaussian moments $\int x^2\phi^2 = 1/(4\pi)$, $\int x^4\phi^2 = 3/(16\pi^2)$ give $\langle\phi,\psi\rangle = 0$. In particular $\|v_d\|_d \to 1$ (indeed it equals $1 + O(h^4)$, using the superalgebraic sampling error for Schwartz functions). Writing $s = \pi x^2$, $q/\pi = s^2/6 - s/4 + 1/32$ has minimum $-1/16$; hence $v_d > 0$ for $h \leq 1$. Normalize $w_d = v_d/\|v_d\|_d$. By (A4), all non-ground eigenvalues stay a fixed positive distance from Λ_d. Orthogonally decomposing $w_d = \alpha_d h_d + z_d$, $z_d \perp h_d$, and using (A6) gives $\|z_d\|_d \leq C'h^4$. Since both w_d and h_d are strictly positive, $\alpha_d > 0$, whence $1 - \alpha_d = O(h^8)$ and

$$\|h_d - w_d\|_d \leq C''h^4. \tag{A7}$$

This is the crucial stronger-than-$O(h)$ approximation; (A4) alone would not suffice at a moving Zak zero.

Define $Z_d b(m,n) = \sum_{r=0}^{d-1} b(m-dr)e^{2\pi i n r/d}$, $0 \le m,n < d$, and $Zf(x,y) = \sum_{r\in\mathbb{Z}} f(x-r)e^{2\pi i r y}$. Unfolding the integer translate in $S_d f$ gives the **exact** identity

$$Z_d S_d f(m,n) = Zf(m/d, n/d). \tag{A8}$$

Cauchy–Schwarz gives $\sup_{m,n}|Z_d e(m,n)| \le \sqrt{d}\,\|e\|_{\ell^2} = d\|e\|_d$. Therefore (A7) and (A8), with $c_d = 1/\|v_d\|_d \to 1$, yield the uniform estimate

$$Z_d h_d(m,n) = c_d\big\{Z\phi(m/d,n/d) + h^2 Z\psi(m/d,n/d)\big\} + O(h^3), \tag{A9}$$

where the $O(h^3)$ constant is independent of m,n. The Zak transform of a fixed Schwartz function is uniformly bounded on the unit square, so the combined discrepancy from $c_d Z\phi$ is $O(h^2)$.

A.3 The Zak zero and its local phase pattern

Here is an elementary product argument for the Gaussian Zak zero. Put $\rho = e^{-\pi}$, $w = e^{2\pi(x+iy)}$. Apart from the nonvanishing factor $2^{1/4}e^{-\pi x^2}$, $Z\phi(x,y)$ equals $L(w) = \sum_{r\in\mathbb{Z}} \rho^{r^2} w^r$. Consider the convergent product on $\mathbb{C}^*$

$$P(w) = \prod_{k=1}^{\infty}(1+\rho^{2k-1}w)(1+\rho^{2k-1}w^{-1}).$$

Both $P(\rho^2 w)$ and $L(\rho^2 w)$ equal $\rho^{-1}w^{-1}$ times their unshifted values. Comparing Laurent coefficients shows the coefficient of w^r in P is $c_0\rho^{r^2}$ (the constant coefficient $c_0 > 0$ because all product coefficients are nonnegative and the empty selection contributes one). Thus $L = P/c_0$, and the product has no zeros except its factor zeros. In $0 \le x,y < 1$, exactly one factor can vanish: $w = -\rho^{-1} = -e^{\pi}$, i.e. $(x,y) = (1/2,1/2)$. Just one factor vanishes there, so the zero is simple. Pairing terms r and $1-r$ also shows $Z\phi(1/2,1/2) = 0$ and

$$A_0 := \partial_x Z\phi(1/2,1/2) \neq 0, \qquad \partial_y Z\phi(1/2,1/2) = iA_0. \tag{A10}$$

For an independent check of nonzero A_0, $A_0 = -4\pi\sum_{k\ge 0}(-1)^k(k+1/2)\phi(k+1/2)$: its summand magnitudes strictly decrease (already the first ratio is $3e^{-2\pi} < 1$), so this alternating series is nonzero. Consequently the real Jacobian at the zero is invertible, and compactness away from it gives a constant $c > 0$ such that $|Z\phi(x,y)| \ge c\sqrt{(x-1/2)^2+(y-1/2)^2}$ on the unit square (reducing c away from the zero). For odd d, every grid point has distance at least $h/\sqrt{2}$ from that zero. Equation (A9) then gives $\min_{m,n}|Z_d h_d(m,n)| \ge c'h - O(h^2) > 0$ for all sufficiently large odd d. This proves eventual nonvanishing, without assuming the conjecture's all-odd-d nonvanishing assertion.

Context and references

The finite Balian–Low theorem of Nitzan and Olsen [2] gives logarithmic growth with unspecified comparison constants in this normalization. The argument here identifies the leading coefficient and compares the finite-Hermite candidate with the unrestricted minimum. Its lower-bound mechanism is a version of the standard growing-ball method for phase vortices [3, 4]; the discrete

magnetic interpolation, seam bookkeeping, and degree argument needed for the Zak problem are included explicitly. Related planar XY and continuum vortex results support the interpretation, but are not silently used as theorems about this twisted discrete torus.

A bounded literature search did not locate a prior theorem with exactly this sharp finite-Zak statement. That does not establish bibliographic novelty. The proof is supplied for independent specialist review; no external peer review or submission is claimed.

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OpenAI Astra assisted with the proof search, verification, and preparation of this note. The argument is written out, including the topological degree lemma and the spectral approximation, so that its conclusions can be checked independently.

References

[1] M. Lammers and S. Stampe, *The finite Balian–Low conjecture*, in: Proc. Int. Conf. Sampling Theory and Applications (SampTA 2015), IEEE, 2015, pp. 139–143. doi:10.1109/SAMPTA.2015.7148867.

[2] S. Nitzan and J.-F. Olsen, *Balian–Low type theorems in finite dimensions*, Math. Ann. **373** (2019), 643–677. arXiv:1707.06449.

[3] R. Alicandro, L. De Luca, A. Garroni, and M. Ponsiglione, *Metastability and dynamics of discrete topological singularities in two dimensions: a* Γ*-convergence approach*, Arch. Ration. Mech. Anal. **214** (2014), 269–330. A sharp discrete-XY theorem from this work is reproduced in the next reference; it is not assumed here as a magnetic-torus lower bound.

[4] M. Cicalese, G. Orlando, and M. Ruf, *The* N*-clock model: variational analysis for fast and slow divergence rates of* N, arXiv:2012.09548, especially the reproduced XY vortex asymptotics in Section 7.